RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 17/12/2025	**CIE – II**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: Enterprise Information Systems		**Course Code : IM355TBB**

Note:
1. Answer all questions

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Define Robotic Process Automation (RPA) and state its primary focus.	2	1	CO1
2.	State the fundamental reason why a company must engage in Business Process Reengineering (BPR) before implementing any major automation initiative.	2	2	CO2
3.	Define the term Data Mart and state how it differs in **scope** from a full Data Warehouse.	2	1	CO3
4.	Name the two foundational processes that are typically executed by Data Mining tools to discover new knowledge within a BI system.	2	2	CO3
5.	A decision-maker uses a BI tool to analyze sales by clicking on a region, then a city, and finally a specific store. Name this OLAP technique and briefly state its purpose.	2	2	CO3
	Part – B			
1	List and explain various methods of business process automation.	10	2	CO2
2	Define Robotic Process Automation (RPA). Explain how it works. List the uses and benefits.	10	2	CO2
3	Explain the core difference between Business Intelligence (BI) and Data Warehousing. Describe why BI is considered “the technology and practice of applying information to make decisions”.	10	2	CO1
4	Describe the four progressive types of Data Analytics used in an EIS. For each type, state the central question it is designed to answer.	10	3	CO3
5	Apply the concept of Online Analytical Processing (OLAP) by explaining its primary function. Illustrate how a retail manager would use an OLAP technique like Drilling Down to analyze a regional sales report.	10	4	CO3

Academic year 2024-2025 (ODD Sem)

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Date	**9th January 2025**	**Maximum Marks**	**50**
Course Code	**IM355TBD**	**Duration**	**90 Min**
Sem	**V**	**CIE – 2**	
ADVANCED DECISION MODELLING (Professional Core Elective)			

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part A			
1.	Mention the situations that make the replacement of items necessary?	02	2	1
2.	What are the three strategies of replacement of items which follow sudden failure mechanisms?	02	2	3
3.	State two differences between preventive maintenance and breakdown maintenance.	02	2	2
4.	Give one advantage and one disadvantage of individual replacement policies.	02	2	2
5.	State the main objectives of using the EOQ model?	02	1	3
	Part B			
1	A firm is considering the replacement of a machine, whose cost price is Rs 12,200, and its scrap value is Rs 200. From experience the running (maintenance and operating) costs are found to be as follows: Year: 1 2 3 4 5 6 7 8 Running Cost (Rs): 200 500 800 1200 1800 2500 3200 4000 When should the machine be replaced?	10	3	1
2	The data on the running costs per year and resale price of equipment A, whose purchase price is Rs 2,00,000 are as follows: Year: 1 2 3 4 5 6 7 Running cost (Rs): 30,000 38,000 46,000 58,000 72,000 90,000 1,10,000 Resale value (Rs): 1,00,000 50,000 25,000 12,000 8,000 8,000 8,000 i. What is the optimum period of replacement? ii. When equipment A is two years old, equipment B, which is a new model for the same usage, is available. The optimum period for replacement is 4 years with an average cost of Rs 72,000. Should equipment A be changed with equipment B? If so, when?	10	3	2

Academic year 2024-2025 (ODD Sem)

<table>
<tr><td>3</td><td>A company is considering the purchase of a new machine at Rs 15,000. The economic life of the machine is expected to be 8 years. The salvage value of the machine at the end of the life will be Rs 3,000. The annual running cost is estimated to be Rs 7,000.
i. Assuming an interest rate of 5 per cent, determine the present worth of future costs of the proposed machine.
ii. Compare the new machine with the presently-owned machine that has an annual operating cost of Rs 5,000 and cost of repair Rs 1,500 in the second year, with an annual increase of Rs 500 in the subsequent years of its life.</td><td>10</td><td>3</td><td>3</td></tr>
<tr><td>4</td><td>The following failure rates have been observed for a certain type of bulbs.
<table>
<tr><td>End of Week:</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr>
<tr><td>Probability of failure to date:</td><td>0.05</td><td>0.13</td><td>0.25</td><td>0.43</td><td>0.68</td><td>0.88</td><td>0.96</td><td>1.0</td></tr>
</table>
The cost of replacing an individual bulb is Rs. 2.25. if the cost of group replacement is 60 paisa per bulb and the total number of bulbs is 1000. What is best interval between group replacements?</td><td>10</td><td>3</td><td>3</td></tr>
<tr><td>5</td><td>Differentiate between Continuous Review and Periodic Review Inventory Systems by the following aspects:
i. Monitoring
ii. Reorder Trigger
iii. Stockouts Risks
iv. System Complexity
v. Suitability</td><td>10</td><td>2</td><td>2</td></tr>
</table>

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars			CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks Distribution	Quiz	Max Marks	1	2	2	–	1	2	2	–	–	–
	Test	Max Marks	10	20	20	–	–	10	40	–	–	–

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RV COLLEGE OF ENGINEERING®, Bengaluru-59

(Autonomous Institution Affiliated to VTU, Belagavi)

Program	CHEMICAL ENGINEERING			
Course Title	GREEN AND HYDROGEN TECHNOLOGY			
Course Code	CH375TGD			
CONTINUOUS INTERNAL EVALUATION - 2				
GLOBAL ELECTIVE: GROUP D				
Sem	V	**Date | Time**	**DEC 2025**	
Max Marks	**60**	**16.12.2025 | 13:20 - 15:20**	**Max Duration**	**120 Min**

Instructions to Candidates:

All the questions are compulsory. No additional booklet will be provided.

Q. No.	QUESTIONS	M	BT	CO
1.1	What are the outcomes of hydrogen risk assessment?	02	01	01
1.2	List the steps involved in the quantitative hydrogen risk assessment.	02	01	04
1.3	Define Codes of Practice.	02	01	03
1.4	What is PAS?	02	01	02
1.5	List the parameters associated with hydrogen for which the codes and standards are defined. Hint: The approval for these parameters vary between countries.	02	01	02
Part B				
2	Explain in detail, how hydrogen gas is stored as compressed gas including its advantages and disadvantages.	10	02	04
3	What are the salient features of storing hydrogen as cryogenic liquid? How is it done and what are the associated benefits and hurdles?	10	02	04
4	Is storing hydrogen as a metal hydride a boon or a bane (annoyance)? Justify with substantial details including the working principle, advantages and disadvantages.	10	03	03
5	How are liquid organic hydrogen carriers beneficial in case of storage and transport of hydrogen? Provide a detailed narration.	10	02	03
6	Give an explanatory note on the codes and standards (national and international) associated with the handling and safety of hydrogen gas.	10	03	01

Course Outcomes

At the end of the course the student will be able to,	
1	Understand the importance of hydrogen and its use as an energy carrier
2	Explain the production, storage and handling of hydrogen
3	Analyze the need for hydrogen as an alternate fuel and the associated challenges
4	Appraise the importance of safety, regulations and codes

Cos/BTL	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks	12	04	22	22	10	30	20	.	.	.